

Estimated Annual Agricultural Pesticide Use Umatilla County, Oregon 2014-2018

24 July 2026

Summary

Umatilla County ranks **#2 of Oregon's 36 counties** by total estimated agricultural pesticide mass applied (EPest-high method, summed 2014-2018), making it one of OR's five highest pesticide-use counties. Umatilla County, adjoining Morrow County, is a major wheat producer and irrigated farming area near Hermiston, growing potatoes, onions, and watermelons.

Table 2 lists the rank and best available estimates of mass applied (kg) for the top 80 agricultural pesticides in **Umatilla County, Oregon** during 2018, as reported by [USGS Estimated Annual Agricultural Pesticide Use](#) data. The table also includes rank by 3-year average (2016-2018) and 5-year average (2014-2018) estimates of mass applied, and each compound's chemical class (abbreviated; see the key in Table 1 and the full classification, with a separate functional class column, in `data/pesticide_classification.csv`, shared with every chapter in this repository). Rankings and estimates are based on the *EPest-high method* described by [Thelin and Stone \(2013\)](#) and [Baker and Stone \(2015\)](#):

“EPest-low and EPest-high, are used to estimate a range of pesticide use. Both EPest-low and EPest-high methods incorporate proprietary surveyed rates for Crop Reporting Districts (CRDs), but EPest-low and EPest-high estimates differ in how they treat situations when a CRD was surveyed and pesticide use was not reported for a particular crop present in the CRD. In these situations, EPest-low assumes zero use in the CRD for that pesticide-by-crop combination. EPest-high, however, treats the unreported use for that pesticide-by-crop combination in the CRD as missing data. In this case, pesticide-by-crop use rates from neighboring CRDs or CRDs within the same region are used to estimate the pesticide-by-crop EPest-high rate for the CRD.” *-Pesticide National Synthesis Project webpage*

USGS lists several caveats of these data, in particular:

- Reliability decreases with scale (e.g. “detailed interpretation of where and how much use occurs within a county is not appropriate”).
- EPest-low estimates are more likely to reflect state-based restrictions on pesticide use than EPest-high estimates.
- A blank cell in the source file (no reported/estimated use for that compound-year-county) is treated as zero in this chapter.
- **2019 is excluded from this ranking.** The 2019 county-level file available at the time of writing covers only 69 compounds nationwide, versus roughly 400 compounds in every other year from 1992-2018 — a partial/preliminary release, not a real drop in the number of pesticides in use. Anchoring the ranking on it would silently omit hundreds of real compounds. This chapter uses 2018 (the most recent complete year) as the 1-year snapshot instead; re-run with 2019 -included windows once a complete 2019 release is available.

Table 1: Chemical class abbreviation key (used in the Class column of Table 2)

Code	Chemical class
CPX	Chlorophenoxy compound
DIP	Dipyridyl compound
DTC	Dithiocarbamates
INO	Inorganic compounds
MIC	Microbial
CB	N-methyl carbamates (AChE inhibitor)
OGM	Organo-metallic compound
OC	Organochlorine compound
OP	Organophosphorous compound

Table 1: Chemical class abbreviation key (used in the Class column of Table 2)
(continued)

Code	Chemical class
OTH	Other
PYR	Pyrethrins/Pyrethroids
TC	Thiocarbamates
TRZ	Triazines

Table 2: Top 80 EPest-high Estimates, Umatilla County, OR, 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above)

Rank	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg
1	DICHLOROPROPENE	OC	216722	230061	247686
2	CHLOROPICRIN	OC	168383	69665	43410
3	METAM	DTC	150160	411079	506371
4	METAM POTASSIUM	DTC	125104	160552	181792
5	BACILLUS AMYLOLIQUEFACIEN	MIC	93288	73507	73507
6	GLYPHOSATE	OTH	71229	63687	76811
7	SULFURIC ACID	INO	63406	142536	138259
8	PARAQUAT	DIP	36672	20942	14636
9	PETROLEUM OIL	OTH	32142	29106	28160
10	BROMOXYNIL	OTH	17413	12435	10268
11	MCPA	CPX	13296	11290	11966
12	DICAMBA	OTH	10787	5701	6179
13	SULFUR	INO	9100	6318	8051
14	2,4-D	CPX	7544	16083	16426
15	PROPICONAZOLE	OTH	6621	4917	3779
16	BURKHOLDERIA SPP	MIC	6367	11132	7430
17	EPTC	TC	5359	15538	22782
18	ATRAZINE	TRZ	4643	3410	2327
19	KAOLIN CLAY	INO	4602	12872	8939
20	DCPA	OTH	4184	5741	5478
21	METRIBUZIN	TRZ	4160	3140	4518
22	OXAMYL	CB	4114	2426	9375
23	PENDIMETHALIN	OTH	3473	3861	4786
24	CLOPYRALID	OTH	3244	3182	2939
25	CHLOROTHALONIL	OTH	3233	8586	9000
26	CALCIUM POLYSULFIDE	INO	3208	3073	2885
27	COPPER	INO	2969	2093	2514
28	AZOXYSTROBIN	OTH	2921	2737	2806
29	CHLORPYRIFOS	OP	2896	2527	2956
30	THIOPHANATE-METHYL	OTH	2876	3396	2593
31	MANCOZEB	DTC	2693	9503	12274
32	MCPB	CPX	2586	1010	878
33	PYRACLOSTROBIN	OTH	2540	2042	2116
34	METOLACHLOR & METOLACHLOR-S	OTH	2537	4584	7143
35	METOLACHLOR-S	OTH	2028	4031	6241
36	METHOMYL	CB	2025	2605	2653
37	DIMETHENAMID & DIMETHENAMID-P	OTH	2015	3425	2849
38	DIMETHENAMID-P	OTH	2015	2599	2353
39	PYRASULFOTOLE	OTH	1446	846	712
40	BOSCALID	OTH	1398	1036	972
41	COPPER HYDROXIDE	INO	1312	3224	2233
42	PYMETROZINE	OTH	1285	663	536
43	TRIFLURALIN	OTH	1284	1309	1135
44	PICOXYSTROBIN	OTH	1252	423	765
45	PYROXASULFONE	OTH	1106	712	1000
46	COPPER OXYCHLORIDE	INO	1015	1029	747

Table 2: Top 80 EPest-high Estimates, Umatilla County, OR, 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above) (*continued*)

Rank	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg
47	PYRIMETHANIL	OTH	1011	1094	823
48	FLUXAPYROXAD	OTH	995	881	619
49	BENTAZONE	OTH	912	496	1586
50	PROPARGITE	OTH	886	444	1155
51	FLUOPYRAM	OTH	823	549	390
52	ALPHA CYPERMETHRIN	PYR	773	397	397
53	CLETHODIM	OTH	742	613	661
54	MALATHION	OP	724	623	561
55	POTASSIUM BICARBONATE	INO	699	735	598
56	MEFENOXAM	OTH	671	803	656
57	ETHOFUMESATE	OTH	608	809	570
58	PYROXSULAM	OTH	552	322	269
59	TRI-ALLATE	TC	545	726	808
60	DIQUAT	DIP	535	719	652
61	IMIDACLOPRID	OTH	518	723	769
62	METOLACHLOR	OTH	509	553	902
63	FLUROXYPYR	OTH	508	2408	2206
64	DIMETHOATE	OP	504	3804	3555
65	BIFENTHRIN	PYR	494	425	356
66	PINOXADEN	OTH	456	422	495
67	CARBARYL	CB	447	314	435
68	FLUTOLANIL	OTH	428	756	862
69	ETHALFLURALIN	OTH	413	1076	1085
70	PENTHIOPYRAD	OTH	385	662	571
71	TERBACIL	OTH	380	190	240
72	CYMOXANIL	OTH	372	533	503
73	BICYCLOPYRONE	OTH	355	210	210
74	GLUFOSINATE	OTH	352	464	340
75	FLONICAMID	OTH	343	343	251
76	TRIASULFURON	OTH	340	201	187
77	TRIFLUMIZOLE	OTH	338	290	317
78	SETHOXYDIM	OTH	334	431	558
79	BACILLUS THURINGIENSIS	MIC	314	246	279
80	FLUCARBAZONE	OTH	295	166	126

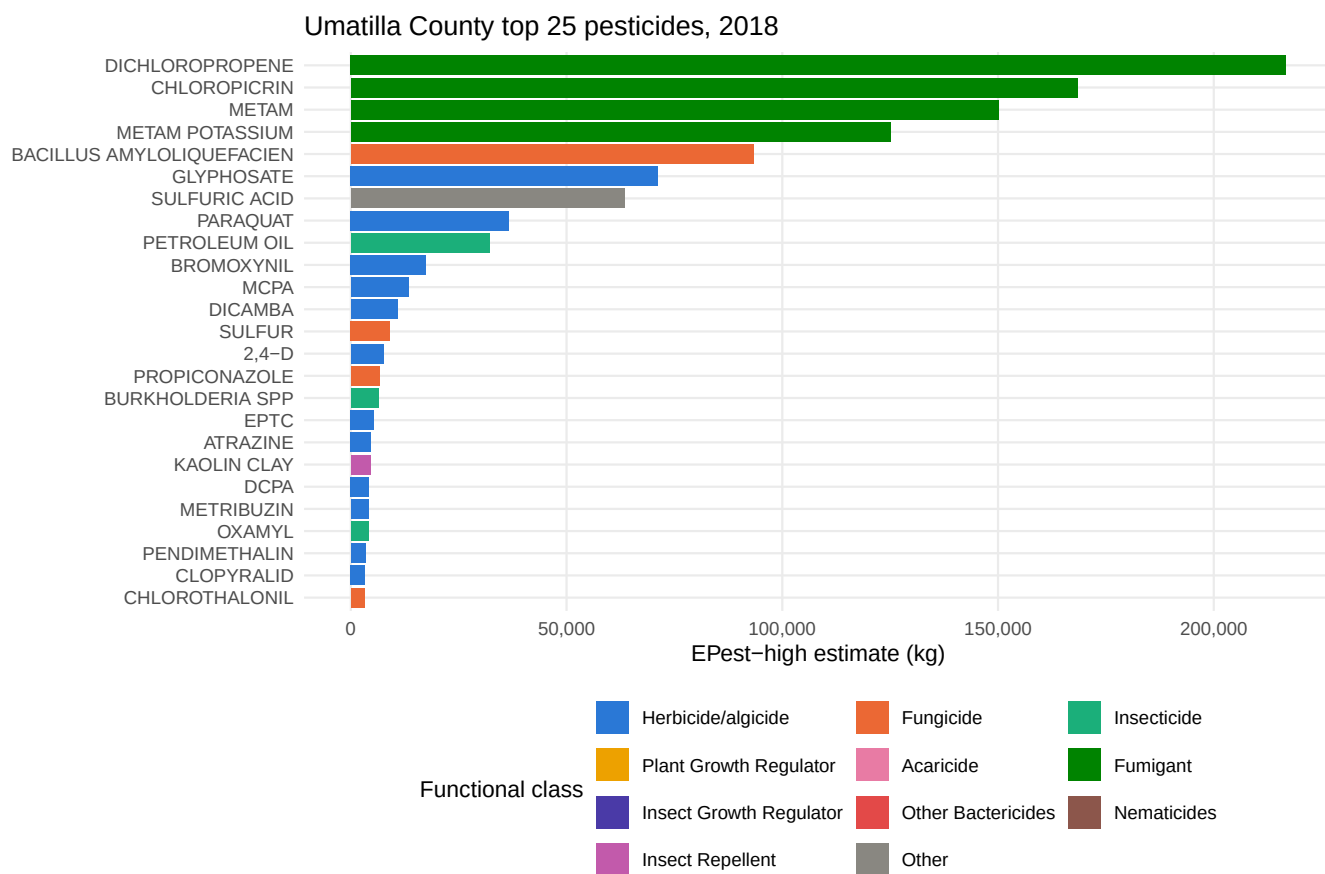


Figure 1: Top 25 compounds in Umatilla County by E Pest-high mass applied, most recent year, colored by pesticide functional class.

Appendices

Table 3: Top 50 Umatilla County Estimates, Range (EPest-low and EPest-high), 1-Yr, 2018

RANK_1YR	COMPOUND	EPEST_LOW_KG_1YR	EPEST_HIGH_KG_1YR
1	DICHLOROPROPENE	216722	216722
2	CHLOROPICRIN	168383	168383
3	METAM	150160	150160
4	METAM POTASSIUM	0	125104
5	BACILLUS AMYLOLIQUEFACIEN	93288	93288
6	GLYPHOSATE	54404	71229
7	SULFURIC ACID	0	63406
8	PARAQUAT	27547	36672
9	PETROLEUM OIL	32142	32142
10	BROMOXYNIL	17146	17413
11	MCPA	13110	13296
12	DICAMBA	10704	10787
13	SULFUR	9100	9100
14	2,4-D	6916	7544
15	PROPICONAZOLE	6621	6621
16	BURKHOLDERIA SPP	6367	6367
17	EPTC	5359	5359
18	ATRAZINE	4643	4643
19	KAOLIN CLAY	4602	4602
20	DCPA	4184	4184
21	METRIBUZIN	1030	4160
22	OXAMYL	4114	4114
23	PENDIMETHALIN	2145	3473
24	CLOPYRALID	0	3244
25	CHLOROTHALONIL	3233	3233
26	CALCIUM POLYSULFIDE	3208	3208
27	COPPER	2969	2969
28	AZOXYSTROBIN	163	2921
29	CHLORPYRIFOS	2382	2896
30	THIOPHANATE-METHYL	718	2876
31	MANCOZEB	2693	2693
32	MCPB	1195	2586
33	PYRACLOSTROBIN	399	2540
34	METOLACHLOR & METOLACHLOR-S	2537	2537
35	METOLACHLOR-S	2028	2028
36	METHOMYL	2025	2025
37	DIMETHENAMID & DIMETHENAMID-P	1909	2015
38	DIMETHENAMID-P	1909	2015
39	PYRASULFOTOLE	1310	1446
40	BOSCALID	1278	1398
41	COPPER HYDROXIDE	1312	1312
42	PYMETROZINE	0	1285
43	TRIFLURALIN	1026	1284
44	PICOXYSTROBIN	0	1252
45	PYROXASULFONE	44	1106
46	COPPER OXYCHLORIDE	1015	1015
47	PYRIMETHANIL	0	1011
48	FLUXAPYROXAD	149	995
49	BENTAZONE	912	912
50	PROPARGITE	169	886

Table 4: Top 50 Umatilla County Estimates, Range (EPest-low and EPest-high), 3-Yr Avg, 2016-2018

RANK_3YR	COMPOUND	EPEST_LOW_KG_3YR_AVG	EPEST_HIGH_KG_3YR_AVG
1	METAM	303850	411079
2	DICHLOROPROPENE	230061	230061
3	METAM POTASSIUM	86910	160552
4	SULFURIC ACID	0	142536
5	BACILLUS AMYLOLIQUEFACIEN	31733	73507
6	CHLOROPICRIN	68581	69665
7	GLYPHOSATE	52732	63687
8	PETROLEUM OIL	29106	29106
9	PARAQUAT	9858	20942
10	2,4-D	7779	16083
11	EPTC	15538	15538
12	KAOLIN CLAY	12872	12872
13	BROMOXYNIL	9410	12435
14	MCPA	8147	11290
15	BURKHOLDERIA SPP	11132	11132
16	ETHOPROPHOS	0	9829
17	PROPAMOCARB HCL	0	9591
18	MANCOZEB	9503	9503
19	CHLOROTHALONIL	8586	8586
20	DIURON	6607	6936
21	SULFUR	6318	6318
22	DCPA	5741	5741
23	DICAMBA	4714	5701
24	PROPICONAZOLE	4846	4917
25	METOLACHLOR & METOLACHLOR-S	3690	4584
26	METOLACHLOR-S	3138	4031
27	PENDIMETHALIN	2748	3861
28	DIMETHOATE	595	3804
29	DIMETHENAMID & DIMETHENAMID-P	3377	3425
30	ATRAZINE	3217	3410
31	THIOPHANATE-METHYL	498	3396
32	PHOSPHORIC ACID	0	3300
33	COPPER HYDROXIDE	1200	3224
34	CLOPYRALID	0	3182
35	METRIBUZIN	1714	3140
36	CALCIUM POLYSULFIDE	3073	3073
37	AZOXYSTROBIN	1753	2737
38	METHOMYL	2324	2605
39	DIMETHENAMID-P	2551	2599
40	CHLORPYRIFOS	2037	2527
41	DIMETHENAMID	2479	2479
42	OXAMYL	1723	2426
43	FLUROXYPYR	288	2408
44	COPPER	2093	2093
45	PYRACLOSTROBIN	1055	2042
46	TRIFLURALIN	822	1309
47	LINURON	187	1286
48	PYRIMETHANIL	757	1094
49	ETHALFLURALIN	685	1076
50	BOSCALID	996	1036

Table 5: Top 50 Umatilla County Estimates, Range (EPest-low and EPest-high), 5-Yr Avg, 2014-2018

RANK_5YR	COMPOUND	EPEST_LOW_KG_5YR_AVG	EPEST_HIGH_KG_5YR_AVG
1	METAM	442033	506371
2	DICHLOROPROPENE	191859	247686
3	METAM POTASSIUM	137607	181792
4	SULFURIC ACID	0	138259
5	GLYPHOSATE	69655	76811
6	BACILLUS AMYLOLIQUEFACIEN	31733	73507
7	CHLOROPICRIN	42755	43410
8	PETROLEUM OIL	28160	28160
9	EPTC	22593	22782
10	2,4-D	9743	16426
11	PARAQUAT	6526	14636
12	MANCOZEB	12174	12274
13	MCPA	9719	11966
14	BROMOXYNIL	8206	10268
15	PHOSPHORIC ACID	8274	9924
16	PROPAMOCARB HCL	0	9591
17	OXAMYL	8953	9375
18	CHLOROTHALONIL	9000	9000
19	KAOLIN CLAY	8886	8939
20	SULFUR	8051	8051
21	BURKHOLDERIA SPP	7430	7430
22	METOLACHLOR & METOLACHLOR-S	5557	7143
23	ETHOPROPHOS	0	6701
24	METOLACHLOR-S	4664	6241
25	DICAMBA	5544	6179
26	DCPA	5478	5478
27	DIURON	4896	5223
28	PENDIMETHALIN	3865	4786
29	METRIBUZIN	3313	4518
30	QUINTOZENE	3846	3937
31	PROPICONAZOLE	3701	3779
32	DIMETHOATE	1286	3555
33	CHLORPYRIFOS	2195	2956
34	CLOPYRALID	3	2939
35	CALCIUM POLYSULFIDE	2885	2885
36	DIMETHENAMID & DIMETHENAMID-P	2795	2849
37	AZOXYSTROBIN	2208	2806
38	METHOMYL	2408	2653
39	METIRAM	2647	2647
40	THIOPHANATE-METHYL	426	2593
41	COPPER	1936	2514
42	DIMETHENAMID	2479	2479
43	DIMETHENAMID-P	2299	2353
44	ATRAZINE	2151	2327
45	COPPER HYDROXIDE	964	2233
46	FLUROXYPYR	481	2206
47	PYRACLOSTROBIN	997	2116
48	DICLOFOP	0	1611
49	BENTAZONE	468	1586
50	PROPARGITE	453	1155